Nonlinear Practice 2

Module 8 - Nonlinear Pharmacokinetics · Daily practice sheet · No date

Name _____ Date _____

1a 1. A 75-kg, 32-year-old female has been taking 200 mg phenytoin daily. Because her average phenytoin plasma concentration was only 6 mg/L, her dose was increased to 350 mg/day. The steady-state average phenytoin concentration was then found to be 21 mg/L. (Volume of distribution = 0.75 L/kg and F = 1) a. Calculate the patient's Vmax and KM. What is the KM?

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1b 1. A 75-kg, 32-year-old female has been taking 200 mg phenytoin daily. Because her average phenytoin plasma concentration was only 6 mg/L, her dose was increased to 350 mg/day. The steady-state average phenytoin concentration was then found to be 21 mg/L. (Volume of distribution = 0.75 L/kg and F = 1) b. Recommend a daily dose of phenytoin for this patient to achieve a phenytoin plasma concentration of 15 mg/L at steady state.

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Question 2 What is the difference between linear and non-linear pharmacokinetics?

A. Nonlinear processes affect absorption only, never elimination
B. Linear parameters stay fixed with dose; nonlinear ones change with dose
C. In nonlinear pharmacokinetics the AUC rises in direct proportion to the dose
D. Linear pharmacokinetics is zero order and nonlinear pharmacokinetics is first order

Question 3 What changes in t½ and Cl_T are usually observed with increased concentrations of drugs that exhibit nonlinear pharmacokinetics?

A. t½ increases and Cl_T increases

B. Neither changes, because Vmax and KM are constants

C. t½ decreases and Cl_T increases

D. t½ increases and Cl_T decreases

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a Her key

Answer: 9 mg/L

WHAT THE STEM GIVES

R1	200 mg	the first daily dose
C_1	6 mg/L	average steady state on 200 mg
R2	350 mg/day	the increased dose
C_2	21 mg/L	steady state on 350 mg/day
weight	75-kg	*not needed for KM*

THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 200 mg/day, C_1 = 6 mg/L, R2 = 350 mg/day, C_2 = 21 mg/L

Two daily doses in the same patient and the average steady-state concentration each produced: the direct method for KM.

ALGEBRA

$$KM = \frac{350 - 200}{(200/6) - (350/21)} = \frac{150}{33.33 - 16.67} = \frac{150}{16.67}$$

Each dose over its own concentration. The 75 kg weight and the 0.75 L/kg volume are not used for KM; they return for the half-life in part c.

ROUND

KM = 9 mg/L

(mg/day) over (mg/day per mg/L) leaves mg/L. 9.00 exactly, because 33.33 − 16.67 is 16.67 and 150 over 16.67 is 9.

> **DOES THE NUMBER MAKE SENSE?**
>
> A 75% rise in dose gave a three-and-a-half-fold rise in concentration, so elimination is saturable; KM is 9 mg/L, a concentration.

1a Her key

Answer: 500 mg/day

WHAT THE STEM GIVES

KM	9 mg/L	from the KM half of part a
R2	350 mg/day	the dose used
C_2	21 mg/L	its own steady state

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax: multiply both sides by (KM + C_{ss}) and divide by C_{ss}, then use either dose with its own steady-state concentration.

ALGEBRA

$$\text{Vmax} = \frac{(350\ \text{mg/day})(9 + 21)\ \text{mg/L}}{21\ \text{mg/L}} = \frac{10\ 500}{21} = 500\ \text{mg/day}$$

mg/L cancels and Vmax keeps the units of the dose. The other pair gives 200 × 15 over 6 = 500 mg/day as well, as it must for one patient.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax must exceed the 350 mg/day dose that reached a steady state: 500 mg/day.

1b Her key

Answer: 315 mg/day

WHAT THE STEM GIVES

KM	9 mg/L	from part a
Vmax	500 mg/day	from part a
C_{ss}	15 mg/L	the target

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{\text{Vmax} \times C_{ss}}{KM + C_{ss}}$$

The patient's own KM and Vmax from part a, with the target of 15 mg/L as C_{ss}, in the nonlinear dose-rate line; no weight is needed because Vmax is in mg/day.

ALGEBRA

$$R = \frac{(500\ \text{mg/day})(15\ \text{mg/L})}{(9 + 15)\ \text{mg/L}} = \frac{7500}{24} = 312.5\ \text{mg/day}$$

mg/L cancels and leaves mg/day. 15 mg/L lies between 6 and 21 mg/L, so the dose lies between the 200 and 350 mg/day she took.

ROUND

$$R = 315\ \text{mg/day}$$

312.5 mg/day rounded up to the nearest 5 mg is 315 mg/day, a dose that can be given; the unrounded 312.5 mg/day is also accepted.

> **DOES THE NUMBER MAKE SENSE?**
> 15 mg/L lies between 6 and 21 mg/L, so the dose lies between 200 and 350 mg/day: 315 mg/day.

Her sheet computes 312.5 mg/day and writes → 315 mg/day; both are accepted.

1c Her key

Answer: 1.87 days

WHAT THE STEM GIVES

weight	75-kg	$V_D = 0.75 \times 75 = 56.25$ L
V_D	0.75 L/kg	times the weight
C_p	15 mg/L	the first stated concentration
KM	9 mg/L	from part a
Vmax	500 mg/day	from part a

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.75 \text{ L/kg} \times 75 \text{ kg} = 56.25 \text{ L}$$

The half-life line needs the volume in litres; the stem gives 0.75 L/kg, so it is multiplied by the 75 kg weight first.

SETUP

$$t½ = V_D\left(\frac{0.693}{Vmax}\right)(KM + C_p)$$

For a saturable drug the half-life depends on the concentration, so it is calculated separately at each stated concentration, here 15 mg/L.

ALGEBRA

$$t½ = 56.25 \text{ L} \times \frac{0.693}{500 \text{ mg/day}} \times (9 + 15) \text{ mg/L} = 1.871 \text{ days}$$

L × mg/L gives mg, which cancels the mg of mg/day and leaves days. Leaving the volume as 0.75 gives 0.025, a result per kilogram.

ROUND

$$t½ = 1.87 \text{ days}$$

1.871 days to two decimal places. At 21 mg/L the same line gives 2.34 days, because KM + C_p is larger and Vmax is unchanged.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax is in mg/day, so the half-life comes out in days: 1.871 days, shorter than at 21 mg/L because KM + C_p is smaller.

1c Her key

Answer: 2.34 days

WHAT THE STEM GIVES

weight	75-kg	$V_D = 56.25$ L
V_D	0.75 L/kg	times the weight
C_p	21 mg/L	the second stated concentration
KM	9 mg/L	from part a
Vmax	500 mg/day	from part a

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.75 \text{ L/kg} \times 75 \text{ kg} = 56.25 \text{ L}$$

The volume per kilogram times the 75 kg weight gives litres, which the half-life line needs; the volume does not change with concentration.

ALGEBRA

$$t½ = 56.25 \text{ L} \times \frac{0.693}{500 \text{ mg/day}} \times (9 + 21) \text{ mg/L} = 2.339 \text{ days}$$

The same line as at 15 mg/L with C_p = 21 mg/L. Only KM + C_p changes, from 24 to 30 mg/L, so the half-life grows by the same ratio.

ROUND

$$t½ = 2.34 \text{ days}$$

2.339 days to two decimal places, longer than the 1.87 days at 15 mg/L: for a nonlinear drug the half-life lengthens as concentration rises.

> **DOES THE NUMBER MAKE SENSE?**
> The half-life at 21 mg/L must be longer than at 15 mg/L, because KM + C_p is larger: 2.339 days.

1c Her key

Answer: 20.8 L/day

WHAT THE STEM GIVES

C_p	15 mg/L	added to KM in the denominator
KM	9 mg/L	from part a
Vmax	500 mg/day	from part a

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}$$

For a saturable drug, clearance depends on the concentration in its denominator, so it is calculated at each stated concentration, here 15 mg/L.

ALGEBRA

$$Cl_T = \frac{500 \text{ mg/day}}{(9 + 15) \text{ mg/L}} = \frac{500}{24} = 20.83 \text{ L/day}$$

mg/day over mg/L leaves L/day, a volume cleared per unit time. The volume of distribution is not needed for clearance.

ROUND

$$Cl_T = 20.8 \text{ L/day}$$

20.83 L/day to one decimal place. At 21 mg/L the clearance is lower, 16.7 L/day, because C_p sits in the denominator.

> **DOES THE NUMBER MAKE SENSE?**

> Clearance at 15 mg/L must be above its value at 21 mg/L, because C_p is in the denominator: 20.83 L/day.

1c Her key

Answer: 16.7 L/day

WHAT THE STEM GIVES

C_p	21 mg/L	added to KM in the denominator
KM	9 mg/L	from part a
Vmax	500 mg/day	from part a

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}$$

The same clearance line as at 15 mg/L, now at the second stated concentration of 21 mg/L, because clearance changes with concentration.

ALGEBRA

$$Cl_T = \frac{500\ mg/day}{(9 + 21)\ mg/L} = \frac{500}{30} = 16.67\ L/day$$

mg/day over mg/L leaves L/day. The larger concentration makes the denominator larger, so the clearance is smaller than the 20.8 L/day at 15 mg/L.

ROUND

$$Cl_T = 16.7\ L/day$$

16.67 L/day to one decimal place. Clearance falls and half-life rises together as concentration rises, because both depend on KM + C_p.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance at 21 mg/L must be below its value at 15 mg/L, because C_p is in the denominator: 16.67 L/day.

Her sheet labels this second clearance "Cl_T 15"; its working uses 21 mg/L.

Question 2 Her key

Answer: B. Linear parameters stay fixed with dose; nonlinear ones change with dose

In linear pharmacokinetics the parameters do not change when different or multiple doses are given, so half-life and clearance stay fixed. In nonlinear pharmacokinetics a larger dose or chronic administration changes them, and a small dose change can give a large concentration change.

Question 3 Her key

Answer: D. t½ increases and Cl_T decreases

$t½ = V_D(0.693/Vmax)(KM + C_p)$ grows as C_p grows, and $Cl_T = Vmax/(KM + C_p)$ shrinks as C_p grows. In one phenytoin patient, going from 15 to 21 mg/L took the half-life from 1.87 to 2.34 days and

clearance from 20.8 to 16.7 L/day.